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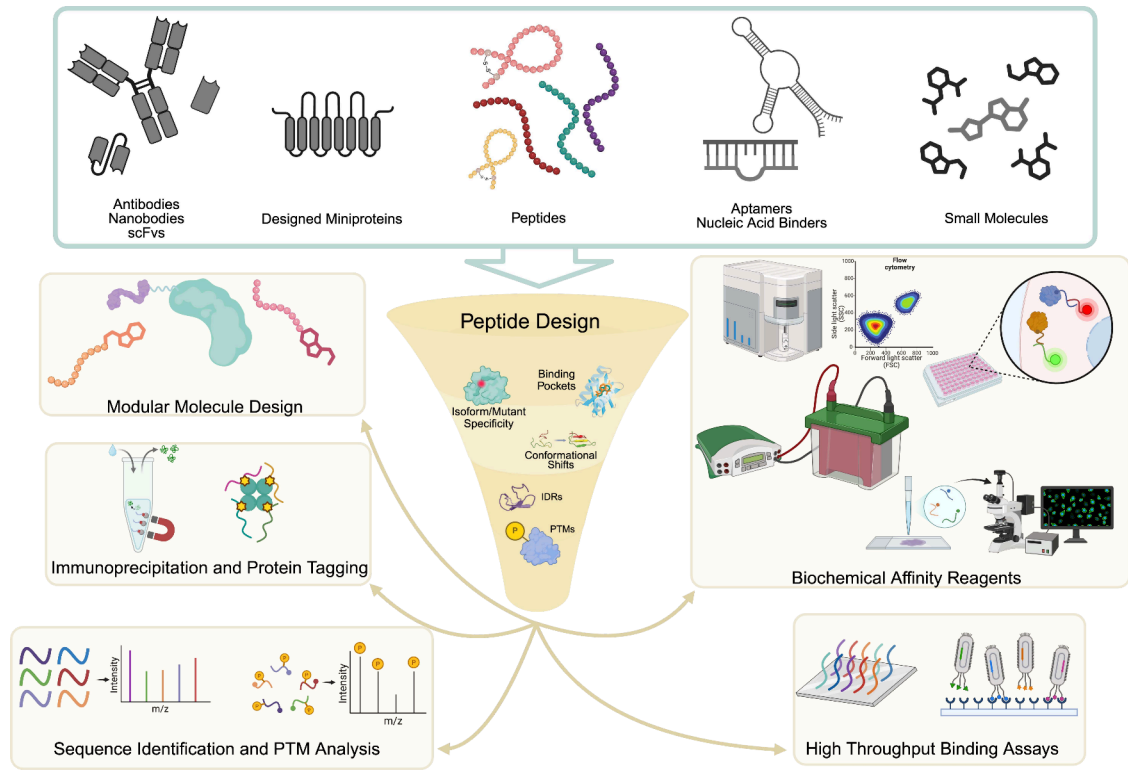


Figure 1. Biochemical Applications of Peptides. Peptides offer unique advantages over other conventional binding modalities for biochemical research, accelerating discovery workflow timelines and broadening the range of targets that are accessible in these landscapes. Created in BioRender. Hong, L. (2026) <https://BioRender.com/c5ttb10>.

Table 1. Summary of Representative Sequence- and Structure-Based Models Used to Embed Peptides or Predict Experimentally Relevant Properties^a

Method	Type	Topology	Alphabet	Predicted Properties
Representation				
ESM-2 ³⁴	Sequence	Linear	WT	Structure prediction (ESMFold), thermostability, PPI prediction, metal ion binding, protein annotation (EC, GO), protein localization ⁶²
PeptideBERT ⁵⁰	Sequence	Linear	WT	Solubility, hemolysis, nonfouling
PepLand ⁵³	Sequence	Both	SMILES	Cell penetration, solubility, binding affinity, synthesizability
HELM-BERT ⁵²	Sequence	Both	HELM	Membrane permeability (cyclic), PPI prediction
PeptideCLM ³⁵	Sequence	Both	SMILES	Membrane diffusion (cyclic)
PepDoRA ⁵¹	Sequence	Both	SMILES	Membrane permeability, nonfouling, hemolysis, PPI prediction
Property prediction				
PepNN-Seq ⁶⁰	Sequence	Linear	WT	Peptide-binding sites on target protein
PepTiVerse ^{6,58}	Sequence	Both	WT, SMILES	Hemolysis, solubility, nonfouling, toxicity, permeability, binding affinity, half-life
PepNN-Struct ⁶⁰	Structure	Linear	WT	Peptide-binding sites on target protein

^aFor representation models, listed properties reflect capabilities demonstrated in the original studies or other cited publications. Topology may be linear, cyclic, or both. See Table S1 for detailed usage notes. ^bCapabilities differ for WT (wild-type) vs SMILES input. ^cThe model accepts peptide sequences alongside target structures.

reagents to a small subset of well-characterized targets (Figure 1).

Artificial intelligence is fundamentally reshaping this landscape. The past several years have witnessed the emergence of machine learning methods capable of designing peptides *de novo* while conditioning on user-specified targets, binding modes, and functional constraints.^{30–33} These approaches leverage advances in language models for both wild-type and noncanonical containing sequences,^{34,35} state-of-the-art generative architectures including diffusion and flow matching,^{36–39} and structure prediction networks^{40,41} to navigate sequence space far more efficiently than traditional screening. Crucially, recent methods have demonstrated that AI-designed

peptides can achieve experimental success rates and affinities competitive with (and in some cases exceeding) those obtained through conventional discovery.^{31,32,42} The result is a shift from discovery-driven to design-driven workflows, in which peptides can be generated on demand and tailored to the specific requirements of a biochemical experiment.

In this Review, we survey AI-designed peptides as practical tools for biochemistry. We specifically focus on experimental utility instead of therapeutic development: we emphasize how peptides can serve as affinity reagents for Western blotting and immunoprecipitation, competitive inhibitors for enzymatic and signaling assays, and functional modules for protein degradation, stabilization, or biophysical interrogation.^{1–3,43–45} We

organize the discussion around the two dominant design paradigms: sequence-based and structure-based, and highlight the classes of peptide tools each paradigm supports. Throughout the article, we aim to provide a framework for selecting AI design methods appropriate to specific experimental goals, anticipating a future in which peptide reagents can be generated on demand to meet the evolving needs of biochemical research.

2. AI DESIGN OF PEPTIDE TOOLS: SEQUENCE-BASED AND STRUCTURE-BASED PARADIGMS

AI algorithms are powerful tools for peptide analysis, design, and refinement. Each core class of AI models—representation models, predictive models, and generative models—provides distinct advantages for biochemistry. Our survey covers both established and emerging computational methods, which were selected to represent the breadth of current peptide design strategies. We include several widely adopted tools as well as newer frameworks that introduce important technical advances and previously unseen design capabilities. Readers should therefore interpret Tables 1–3 as a map of the current methodological landscape, rather than as a list of uniformly mature or equally validated tools.

Representation Learning

Representation models produce high-dimensional numerical vectors, or embeddings, that intrinsically capture the defining physicochemical and structural properties of molecular sequences. They learn from data sets containing thousands to millions of sequences by recovering hidden segments via masked language modeling (MLM). Protein language models (pLMs)³⁴ such as ESM-2, ProtT5, and ProteinBERT are trained on full-length proteins,^{34,46,47} and they extend naturally to peptides composed of the 20 canonical amino acids. Chemical language models (CLMs) including ChemBERTa and Chemformer, which are trained on fully flexible chemical notation such as SMILES strings, can potentially represent any natural, chemically modified, or cyclized peptide.^{48,49} Researchers have also developed pepLMs tailored specifically to peptides. This is accomplished either by fine-tuning existing pLMs or CLMs,^{50,51} such as PeptideBERT and PepDoRA, or by training encoder architectures from scratch,^{35,52} such as HELM-BERT and PeptideCLM (Tables 1 and S1). While PeptideBERT⁵⁰ maintains the standard natural amino acid vocabulary, several pepLMs such as PeptideCLM, PepDoRA, PepLand, and HELM-GPT instead utilize SMILES^{35,51,53} and HELM⁵⁴ notation, enforcing a valid peptide backbone while allowing for limitless diversity in the variable groups.

Most modern representation models, both for peptides and the broader field of natural language processing, use Transformer⁵⁵ architectures such as BERT.^{34,35,50–52,56} The Transformer's self-attention mechanism strengthens pepLMs by modeling how each residue or chemical token interacts with the rest of the peptide, capturing long-range and context-dependent effects. Graph-based modeling is another attractive approach, given its intuitive characterization of peptides as 2D molecular graphs, and is utilized by PepLand.⁵³ Pretrained representation models are crucial for downstream sequence-based modeling, streamlining training and consistently improving performance for both predictive and generative algorithms.^{52,34,50,57,58} In contrast, structure-based models typically learn task-specific structural representations end-to-

end, jointly with their folding, prediction, or design objective.^{40,59–61}

Property Prediction

Property prediction is often used as a benchmark in representation learning; the accurate prediction of physicochemical characteristics from peptide embeddings validates the information-rich nature of the learned representations (Tables 1 and S1). Standalone predictive models are also valuable tools in their own right, estimating how a peptide may behave in the context of a target protein or the broader cellular environment. Several models including DELPHI, MEG-PPIs, and EGRET have been trained to predict protein–protein interaction sites.^{63–65} PepCNN and PepBCL predict generic peptide-binding regions on target proteins,^{60,66,67} while PepNN provides one sequence-based (PepNN-Seq) and one structure-based (PepNN-Struct) approach to target interaction hotspot prediction given a specific peptide partner.⁶⁰ These models help elucidate binding mechanisms and guide the design of motif-specific binders. In addition, many pharmacologically and experimentally relevant properties now have sufficient public data to support predictive modeling, including solubility, hemolysis, cell penetration, membrane permeability, synthesizability, nonfouling behavior, binding affinity, half-life, and toxicity. PeptiVerse trains state of the art predictors for several therapeutically relevant properties⁵⁸ (Tables 1 and S1).

Generative Algorithms

AI methodologies for designing peptide sequences and structures, including fully *de novo* design, editing, and scaffolding, fall under the umbrella of generative modeling. In this setting, a model is trained to sample new peptide sequences from a learned distribution that approximates a target distribution of interest.⁶⁸ Depending on the application, this target distribution may correspond to all naturalistic peptides,⁶⁹ cyclic or chemically modified peptides,⁵⁴ peptides that bind a specific protein target,^{12,32} peptides engaging defined hotspot residues,^{12,70} or peptides that simultaneously satisfy binding and physicochemical constraints such as solubility or stability.^{30,71}

Early work in peptide generation relied heavily on autoregressive models, which adopt the Transformer-based GPT architecture⁷² and generate sequences token by token. Autoregressive models have been successfully applied to protein and peptide design,^{54,73–75} but their left-to-right generation precludes the use of bidirectional context and complicates objective-guided sampling. In practice, this makes it difficult to incorporate constraints that depend on global sequence properties or downstream experimental readouts.

More recent approaches therefore favor the bidirectional diffusion- and flow-based generative frameworks, which support flexible conditioning and multi-objective guidance. In discrete diffusion models for sequences, generation is framed as the reversal of a gradual corruption process where tokens are randomly masked or replaced over time. To transform the corrupted sequences back into clean sequences from the data distribution, discrete diffusion models perform individual token transitions across many small time steps.^{37,68,76} These models operate naturally on discrete alphabets, making them well-suited to peptide sequences and SMILES representations. Flow matching provides a closely related but distinct perspective, learning a time-dependent vector field that transports samples from a simple base distribution (e.g., random or corrupted data) to the data distribution consisting of real samples.^{39,77,78}

Table 2. Structure-Based AI Algorithms for Peptide Folding and Design^a

Method	Topology	Alphabet	Conditioning	Experiments
Structure prediction				
AlphaFold2 ⁴⁰	Linear	WT	-	-
AlphaFold2-Multimer ⁸¹	Linear	WT	-	-
RoseTTAFold All-Atom ⁸²	Linear	WT	-	-
AlphaFold3 ⁴¹	Linear	WT	-	-
Ensemble prediction				
PepFlow ⁸⁴	Both	WT	-	-
BioEmu ⁸⁵	Linear	WT	-	-
Peptide design^b				
HelixDiff ⁸⁷	Linear (L- or D-chirality)	WT	Scaffold residue indices	SS
AfCycDesign ⁹⁰	Cyclic	WT	Scaffold structure	B,S
Peptide binder design^c				
RFdiffusion ¹²	Linear	WT	Target structure, binding sites	B,S
RFpeptides ³¹	Cyclic	WT	Target structure, binding sites	B,S
EvoBind2 ⁴²	Both	WT	Target sequence	B
BindCraft ¹¹	Linear	WT	Target structure, binding sites	B,C,S
BoltzGen ⁵⁹	Both	WT	Target structure, covalent bonds, binding sites, residue constraints	B,C
HelixFlow ⁸⁸	Linear (L- or D-chirality)	WT	Scaffold residue indices, target structure	B,C,SS ^d
CP-Composer ⁶¹	Cyclic	WT	Target structure, geometric constraints (e.g., disulfide bonds)	-
CpSDE ⁸⁶	Cyclic	WT	Target structure, cyclization bond graph	-
CYC_BUILDER ⁸⁹	Cyclic	WT	Target structure, binding sites, seed fragments	B,C,SS
PXDDesign ⁹²	Linear	WT	Target structure, binding sites	B

^aConditioning options are not necessarily applied together, or at all (see Table S2 for detailed usage notes). Wet lab experiments validating any model capability (some features may lack validation) are indicated (B = biophysical assay, C = *in cellulo* functional assay, S = 3D structure, SS = secondary structure, V = *in vivo*). Topology may be linear, cyclic, or both. WT = wild-type. ^bModels do not explicitly accept target-related input but could potentially be leveraged for binder design through choice of scaffold, fine-tuning, etc. ^cModels are specifically formulated for binder design, although several can be used without target-related input to generate standalone peptides. ^dTarget-conditioned generation capabilities were not experimentally validated.

Discrete variants of flow matching extend this idea to sequence spaces, enabling a smooth transport over categorical states. In practice, this can enable faster sampling and more globally coordinated sequence updates, although these models can be more complex to implement and train.

Across both diffusion and flow matching, the learned generative model captures the overall structure of the peptide distribution. Additional constraints, such as target specificity and property optimization, can be later introduced through conditional and multi-objective extensions, such as reinforcement learning and Monte Carlo Tree Search, which explicitly bias sampling toward sequences that satisfy competing constraints.^{30,79} This separation of learning a broad peptide prior from steering generation toward task-specific regions of sequence space has proven particularly effective for designing peptides that must balance binding, selectivity, and physicochemical behavior.

Structure-Based Peptide Design

Folding and Conformational Sampling Algorithms. Structure-based design represents one of the earliest rational approaches to protein and peptide engineering, and was among the first paradigms to benefit from modern AI. This strategy is based on the idea that functional polypeptides can be understood and designed directly from their three-dimensional structures. Decades of structural biology have produced an extensive repository of protein structures,⁸⁰ both as monomers and in complex with other proteins, ligands, and peptides. These structures provide atomic-resolution templates for identifying binding motifs, interaction hotspots, and geometric constraints. More recently, structure prediction or “folding” algorithms, including AlphaFold2, AlphaFold-Multimer, Alpha-

Fold3, and RoseTTAFold All-Atom,^{34,40,41,81–83} have reached exceptionally high accuracy on conformationally stable proteins. These tools enable cheap structural analyses of *de novo* designs and structure-based binder design for targets with no solved structure (Tables 2 and S2). Alongside single-structure predictors, ensemble and dynamics-oriented methods such as PepFlow and BioEmu model conformational variability in peptides and proteins,^{84,85} providing a richer view of accessible binding geometries (Tables 2 and S2).

Three-Dimensional Peptide Design. Building on these advances, a new generation of AI-driven methods has enabled peptide design directly in three-dimensional space (Tables 2 and S2). Models such as CpSDE, CP-Composer, CYC_BUILDER, PXDesign, BoltzGen, HelixDiff, and HelixFlow output precise atomic coordinates alongside sequences. These tools either perform sequence-structure codesign^{59,61,86–89} or inverse-fold the corresponding peptide sequence post hoc using structure-to-sequence tools such as ProteinMPNN.^{12,31,90–92} Several approaches focus on general peptide design by scaffolding fixed sequence motifs or performing structure-conditioned redesign, producing either linear^{87,88} or cyclic⁹⁰ peptides. The hotspot inpainting modules of HelixFlow and HelixDiff^{87,88} can optionally incorporate a mirror transformation to produce D-amino acids in the final structure, which is recommended to improve stability and reduce immunogenicity. By selecting peptide template residues known to promote interaction with a protein of interest, users can exploit these models for binder redesign. Meanwhile, AfCycDesign can hallucinate cyclic peptides *de novo* or perform structure-conditioned redesign. When combined with motif grafting tools, AfCycDesign can ultimately yield

Table 3. Sequence-Based AI Algorithms for Naturalistic and Target-Binding Peptide Design^a

Method	Topology	Alphabet	Conditioning	Experiments
Multi-objective peptide design^b				
HELM-GPT ⁵⁴	Cyclic	HELM	Reinforcement learning with properties (permeability, KRAS affinity)*	-
PepThink-R1 ⁹⁹	Cyclic	SMILES	LogD, MRT, SIF	-
PepINVENT ¹⁰⁰	Both	CHUCKLES (WT, noncanonicals)	Template, topological constraints, solubility, permeability	-
Peptide binder design^c				
SalT&PepPr ⁴⁴	Linear	WT	Binding protein seq	B,C,V ¹⁰¹
PepMLM ³²	Linear	WT	Target protein seq	B,C
PepPrCLIP ⁶⁹	Linear	WT	Target protein seq	B,C
Gumbel-FM ¹⁰²	Linear	WT	Target protein seq	-
Metalorian ⁹⁶	Linear	WT	Target metals (e.g., Cu)	B
Multi-objective peptide binder design^c				
PepTune ³⁰	Both	SMILES	Target protein seq (affinity), permeability, solubility, hemolysis, nonfouling	-
moPPIt ⁷⁰	Linear	WT	Target protein seq (affinity), binding sites, half-life, solubility, hemolysis, nonfouling	B,C
MOG-DFM ¹⁰³	Linear	WT	Target protein seq (affinity), half-life, solubility, hemolysis, nonfouling	See moPPIt
AReURDi ¹⁰⁴	Both	WT, SMILES	Target protein seq (affinity), half-life, solubility, hemolysis, nonfouling	-
TR2-D2 ⁷⁹	Both	SMILES	Target protein seq (affinity), permeability, solubility, hemolysis, nonfouling	-
SOAPIA ⁷¹	Linear	WT	Target protein seq (affinity), off-target protein seqs (specificity)	-

^aConditioning options are not necessarily applied together, or at all (see Table S3 for detailed usage notes). Wet lab experiments validating any model capability (some features may lack validation) are indicated (B = biophysical assay, C = *in cellulo* functional assay, S = 3D structure, SS = secondary structure, V = *in vivo*). We define properties: LogD = distribution coefficient (lipophilicity), MRT = mean residence time, SIF = scaffold interaction fingerprint. Topology may be linear, cyclic, or both. WT = wild-type. ^bModels do not explicitly accept target-related input but could potentially be leveraged for binder design through a sequence template, fine-tuning, etc. ^cModels are specifically formulated for binder design, though several can be used without target-related input to generate standalone peptides. *Feature requires additional training.

binding peptides.⁹⁰ However, since the binder design capabilities of HelixDiff and AfCycDesign^{87,88,90} are not presented as an integrated pipeline, instead requiring rational design steps and occasionally external software, we do not yet classify these tools as binder design algorithms (Tables 2 and S2).

Other methods explicitly condition design on a target protein, such as CpSDE, EvoBind2, and HelixFlow's PocketFlow module.^{42,86,88} Providing finer control over the design objective, several tools can incorporate a particular surface or hotspot region within the receptor structure, including RFdiffusion, RFpeptides, PXDesign, CYC BUILDER, CP-Composer, BindCraft, and BoltzGen^{11,12,31,59,61,89,92} (Tables 2 and S2). The structure-based approach excels when high-quality structural information is available, precise binding geometries are required, and the target protein has a dominant binding interface. For the biochemist, these methods can accelerate design and improve the performance of high-affinity binders in pulldown, biophysical, and *in vitro* binding assays. However, reliance on structure is inherently limiting to targets and interactions that can be described in structural space. Sequence-based generative models, by contrast, operate directly in sequence space and can explore a broader range of targets, interaction modes, and chemical modifications, including cases where structural data are sparse or uncertain.⁹³

Sequence-Based Peptide Design

In contrast to structure-based methods, which focus solely on local geometric interaction constrained by conformations captured in the PDB, sequence-based peptide design leverages amino acid sequence to model binding, structure, and physicochemical behavior across a broader and more dynamic landscape (Tables 3 and S3). This view has proven especially effective for targets governed by short linear motifs, flexible interaction surfaces, or intrinsically disordered regions. Early work relied on extracting linear binding segments from known

interaction partners, as in SalT&PepPr,⁴⁴ which uses ESM-2-based representations³⁴ to identify and repurpose sequence fragments from interacting proteins as peptide binders. Moving to *de novo* binder discovery, PepPrCLIP⁶⁹ introduced a design framework inspired by the contrastive-language image pretraining (CLIP) framework of OpenAI's DALLIE-2 model,^{94,95} enabling virtual screening of pLM-sampled peptide libraries directly against target protein sequences. Finally, PepMLM removed the screening step altogether by generating peptide sequences conditioned directly on the target protein using a span masked language modeling objective.³² Together, these pLM-based approaches (Tables 3 and S3) have been experimentally validated across diverse and challenging targets, including mutant Huntington protein, transcription factors, viral phosphoproteins, and oncogenic fusion proteins, demonstrating that reliable peptide binders can be designed from sequence alone even in structurally heterogeneous settings.

Specificity-Constrained Sequence Design. For many biochemical applications, however, affinity alone is insufficient, motivating sequence-based methods that encode biological specificity directly into generation (Tables 3 and S3). moPPIt, which integrates a PepNN-Seq-inspired binding-site predictor, biases peptide generation toward user-defined regions of the target sequence, enabling generation of epitope-specific peptides that can serve as both inhibitors and binding moieties for cell therapies.⁷⁰ Isoform- and paralog-level discrimination further becomes possible when off-target sequences are incorporated explicitly during training, an approach taken by SOAPIA to reduce cross-reactivity in the absence of structural data.⁷¹ Sequence-based design has also been extended beyond protein recognition. As a representative example, latent diffusion approaches such as Metalorian generate peptides that selectively bind and sequester specific metal ions, illustrating that sequence-only frameworks can encode non-protein binding chemistry.⁹⁶ Looking forward, more specialized sequence representations, including those tailored to

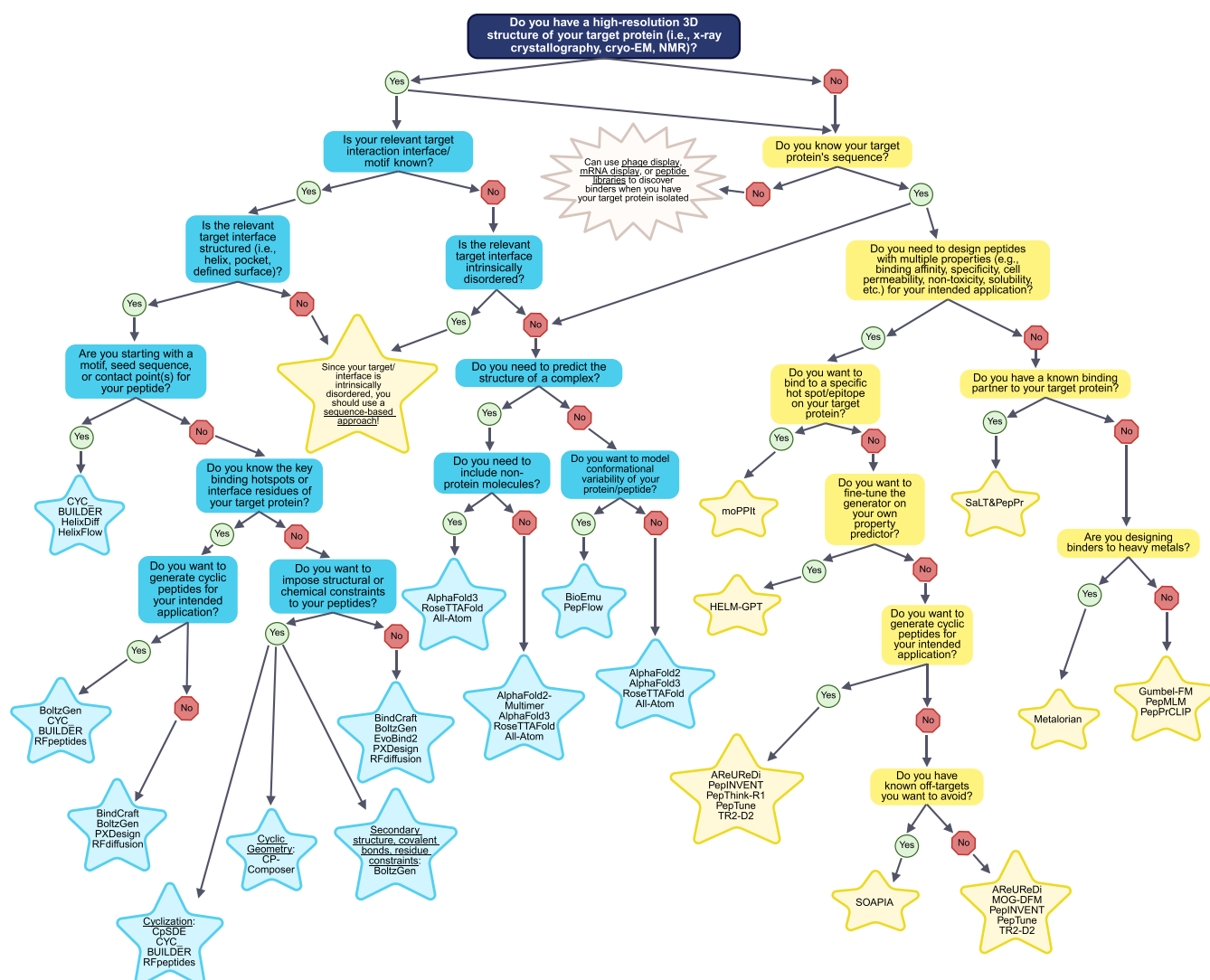


Figure 2. Decision Tree for Structure- and Sequence-Based peptide Design. An experimentalist-driven decision guide for structure- (blue) and sequence-based (yellow) peptide design algorithms. *Note:* Even with resolved 3D structures, sequence-based algorithms are still usable given a known sequence of the target protein. AlphaFold-predicted structures can be used as an input for structure-based algorithms, though experimentally validated structures are more ideal. Created in BioRender. Hong, L. (2026) <https://BioRender.com/cS8tb10>.

post-translationally modified proteins⁹⁷ or specific protein classes,⁹⁸ will further expand the range of selective interactions that can be addressed with peptides without reliance on structural information.

Multi-objective Peptide Generation. Sequence-based peptide design becomes substantially more powerful when combined with flexible chemical languages such as SMILES, HELM, and CHUCKLES, which support noncanonical amino acids, chemical modifications, and macrocyclic topologies (Tables 3 and S3). For example, HELM-GPT⁵⁴ operates directly in the HELM formalism to generate macrocyclic peptides and applies reinforcement learning to optimize properties such as membrane permeability, while PepThink-R1⁹⁹ combines SMILES representations with large language model reasoning to generate cyclic peptides with favorable ADMET profiles. PepINVENT, a Transformer-based reinforcement learning approach, was trained on a vast virtual library of synthesizable peptides (many of which included noncanonical amino acids) using a SMILES-compatible representation called CHUCKLES. The model supports design

of naturalistic peptides and peptidomimetics, as well as peptide property optimization.¹⁰⁰

Newer multi-objective frameworks build specifically on discrete diffusion models^{37,76} trained over peptide SMILES, a powerful linear chemical representation.^{35,105} For example, PepTune³⁰ and TR2-D2⁷⁹ both use a peptide SMILES-based masked discrete diffusion model (PepMDLM) as a base generator, in which sequences are sampled by reversing a corruption process. In PepTune, Monte Carlo Tree Search evaluates candidate sequence edits using pretrained property predictors and steers generation toward sequences that jointly satisfy binding and physicochemical constraints,³⁰ while TR2-D2 introduces reinforcement learning by exponentially tilting the base diffusion distribution toward high-reward trajectories.⁷⁹

A complementary class of methods is based on discrete flow matching.^{39,77,78,102} As an example, MOG-DFM¹⁰³ learns a time-dependent transport field that evolves sequences toward the Pareto front across user-defined objectives and serves as the generative backbone for moPPIt.⁷⁰ A newer variant, AReURDi, extends discrete flow matching through rectified

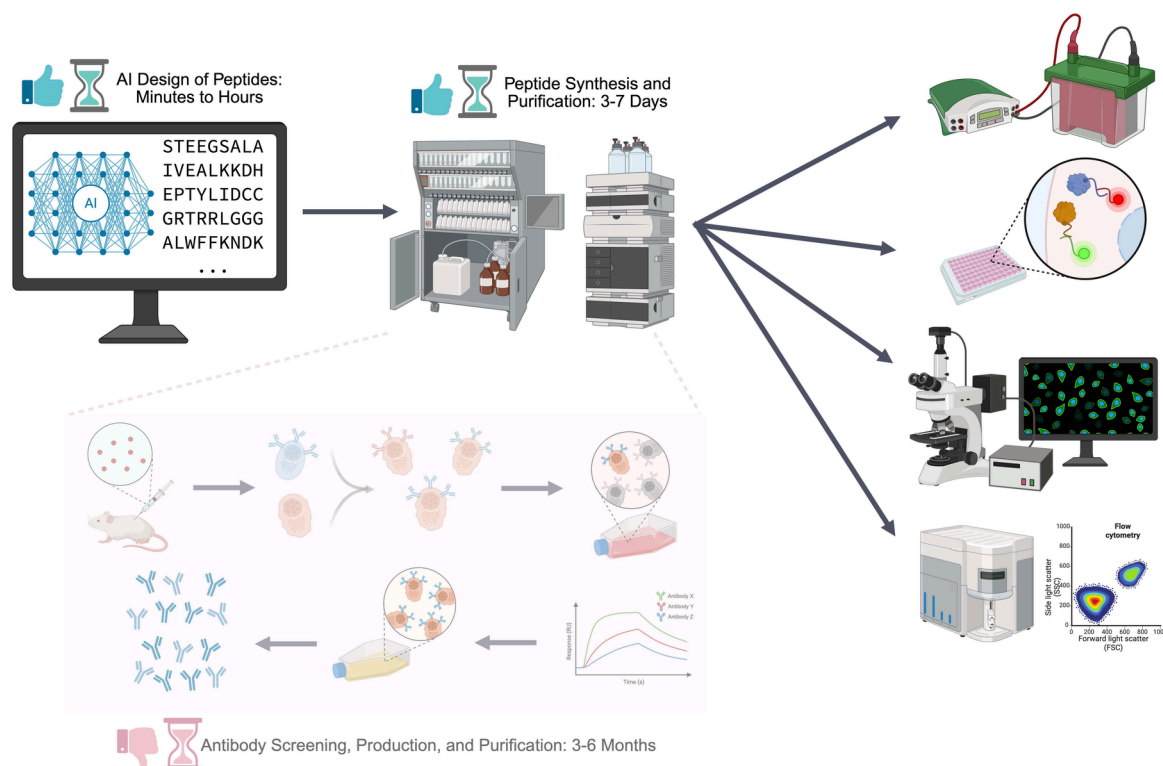


Figure 3. Manufacturing of AI-designed Peptides for Biochemical Affinity Reagents. AI-based design and automated peptide synthesis alleviates manufacturing and production timelines and labors, enabling more programmable, multi-objective peptide generation for biochemical affinity-based applications. Created in BioRender. Hong, L. (2026) <https://BioRender.com/c5ttb10>.

updates at the residue level, enabling more direct movement toward the Pareto front via targeted sequence transitions.¹⁰⁴ Across diffusion-, flow-, and reinforcement-learning-based frameworks (Tables 3 and S3), effective multi-objective control depends critically on accurate pretrained property predictors,^{50,58} which are used either to guide inference-time sampling or to define reward functions for fine-tuning the base generative model. This reliance further emphasizes the importance of close collaboration with experimentalists to generate high-quality biochemical and physicochemical training data, enabling the design of increasingly capable biochemical tools.

Although not comprehensive, particularly given the rapid pace of model development in binder design, we provide a decision tree that suggests an experimentally facing framework to select between structure- and sequence-based peptide design approaches (Figure 2).

3. PEPTIDE TOOLS FOR BIOCHEMICAL ASSAYS AND PROTEIN INTERROGATION

Peptides as Assay-Matched Affinity Reagents

A key advantage of peptides as affinity reagents is their adaptability across assays that place very different demands on binding (Figure 3). Antibodies are commonly reused across Western blotting, immunoprecipitation, and flow cytometry, even though the target protein is presented in fundamentally different states in each case.^{1–3,43} This practice often leads to inconsistent performance that cannot be corrected without completely switching reagents,⁴³ a challenge compounded by the labor-intensive screening and production processes of both polyclonal and monoclonal antibodies, which require immunization of host organisms and maturation of hybridoma cells

(Figure 3).¹⁶ Peptides, which are easily synthesized and readily scalable for high-throughput assays (Figure 3), avoid this problem because peptide binders can be generated with assay conditions specified at the outset.

Linear sequence segments that remain exposed after denaturation can be targeted for Western-blot detection. Native interaction surfaces or short binding motifs (e.g., HaloTags,²¹ poly-Histidine tags,¹⁰⁶ and HiBiT tagging^{107,108}) can be selected for enrichment in immunoprecipitation and pull-down experiments using detergent-containing lysates. Extracellular loops and domains can be targeted for surface staining and flow cytometry,¹⁰⁹ with sequence-level control over charge and hydrophobicity to limit nonspecific membrane association. Because peptide sequences are fully defined and chemically synthesized, the binding strength, background, and labeling chemistry can be adjusted directly when an assay underperforms.

Intracellular Target Engagement and Live-Cell Measurements

Moving beyond lysate-based measurements, the small size of peptides becomes a major advantage. While antibodies generally require fixation, permeabilization, or cell lysis to access intracellular targets, steps that disrupt localization and transient interactions,^{2,16} peptides, by contrast, can be designed to enter cells and remain stable long enough to engage intracellular proteins.^{110,111} This enables the direct measurement of target engagement, binding site occupancy, or competitive displacement in intact cells. Furthermore, by fluorescently labeling peptide binders, intracellular flow-based or imaging readouts^{109,111} become simpler, while unlabeled peptides can be used in competition experiments followed by standard biochemical analysis.^{17,23,112} These workflows pre-

serve cellular organization and allow binding and perturbation to be assessed under conditions closer to those of the native state of the system.

Peptide-Based Probes and Reporters

Peptides also underpin a wide range of biochemical probes, where improved design directly sharpens specificity and interpretability.^{16,27,113} Protease assays typically rely on fluorogenic peptide substrates,^{111,114} while kinase and phosphatase assays use short peptide substrates or docking motifs.¹¹⁵ Consensus sequences are convenient but often report on enzyme families, rather than individual activities. Carefully designed peptide substrates allow selectivity to be shifted toward specific enzymes or closely related isoforms, improving the resolution in complex samples.

Beyond substrates, peptides conjugated to fluorophores,¹¹¹ quenchers,¹¹⁶ FRET pairs,^{114,117} photo-cross-linkers,¹¹⁸ or proximity-labeling handles form the basis of reporters that track binding, cleavage, conformational change, or transient interactions.¹⁰⁸ In all of these cases, sequence context determines the background signal, dynamic range, and off-target effects, making peptide design a central determinant of assay quality.

Programmable Proteome Editing via Induced Proximity

Perhaps the most intriguing opportunity for peptides as biochemical tools is the degree of programmability they introduce at the protein level (Figures 1 and 3). Genome editing technologies such as CRISPR fundamentally transformed biology by making genome manipulation simple, programmable, and widely accessible. Users can specify a guide RNA complementary to a target DNA sequence and recruit a Cas9 nuclease to that site to perform editing.^{119,120} This separation of targeting and action have made it straightforward to knock out, insert, repress, or activate genes across diverse systems.¹²¹ What if the same level of programmability were possible at the protein level? Peptides have helped move toward this goal by allowing users to target specific proteins and recruit enzymatic activity to them. As key examples, genetically encoded peptide-guided E3 ubiquitin ligases (uAbs)^{32,44,69,122} use “guide” peptides fused to E3 ligase catalytic domains to direct ubiquitination and degradation of selected proteins, while peptide-directed deubiquitinases (duAbs)⁴⁵ similarly use peptides fused to the OTUB1 deubiquitinase domain to stabilize or otherwise modulate protein abundance. This programmable architecture enables direct manipulation of protein abundance and regulation without altering the underlying DNA sequence.

Related strategies, including PepTACs,¹²³ DUBTACs,^{124,125} DEPTACs,^{126,127} and other peptide-guided recruitment systems,^{113,128,129} extend this framework to additional enzymatic activities, although most current implementations rely on fusion proteins rather than fully synthetic bifunctional peptides. As peptide binders become easier to design with defined affinity, specificity, and intracellular compatibility, these approaches provide a programmable means of perturbing the proteome that complements genome editing and enables systematic control over protein turnover, interaction networks, and post-translational modifications using standard biochemical assays.

4. OUTLOOK

At present, the most immediate contribution of AI-designed peptides to biochemistry is the ability to generate target-

binding reagents that are explicitly optimized for experimental use via multi-objective peptide generation.^{30,79,103,104} In practice, this provides a compelling alternative to reliance on off-the-shelf antibodies, whose performance is often context-dependent and difficult to predict across different assays, cell types, or buffer conditions.⁴ These innovations have shifted peptide discovery from a sequential, trial-and-error process toward a more engineered workflow, in which peptides are produced as assay-ready reagents rather than minimally functional binders that require extensive downstream refinement.¹³⁰ As a result, peptides can increasingly serve as practical affinity reagents, competitive inhibitors, and modular components for biochemical assays, especially for targets that have historically been difficult to address or reproducibly bind with antibodies or small molecules.^{4,5}

Looking forward, the same design principles are likely to extend beyond target binding toward functional perturbation and cellular context. Emerging models that couple molecule treatment to downstream biochemical or cell-state effects suggest a future in which peptides can not only serve as targeted binders, but also as inducers of defined biochemical outcomes, such as degradation, stabilization, or pathway modulation.^{32,45,131–135} Furthermore, as AI-driven design is integrated with automated peptide synthesis,¹³⁶ high-throughput screening,¹³⁷ and closed-loop experimental feedback,^{138,139} peptide reagents will be generated, screened, and refined on demand. In this regime, we envision that peptides move beyond simple affinity tools and become programmable molecular reagents that link sequence design directly to biochemical function and cellular behavior.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.biochem.6c00138>.

Tables S1, S2, and S3 provide practical guidance to run tools described in main text Tables 1, 2, and 3 (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Pranam Chatterjee – Department of Bioengineering and Department of Computer and Information Science, University of Pennsylvania, Philadelphia, Pennsylvania 19104, United States; orcid.org/0000-0003-3957-8478; Email: pranam@seas.upenn.edu

Authors

Lauren Hong – Department of Bioengineering, University of Pennsylvania, Philadelphia, Pennsylvania 19104, United States

Sophia Vincoff – Department of Bioengineering, University of Pennsylvania, Philadelphia, Pennsylvania 19104, United States

Complete contact information is available at: <https://pubs.acs.org/10.1021/acs.biochem.6c00138>

Author Contributions

[§]L.H. and S.V. contributed equally to this work.

Notes

The authors declare the following competing financial interest(s): P.C. is a co-founder of Gameto, Inc., UbiquiTx,

Inc., Atom Bioworks, Inc., and Recognition Bio, Inc., and advises companies involved in peptide therapeutics development. P.C.'s interests are reviewed and managed by the University of Pennsylvania in accordance with their conflict-of-interest policies. The remaining authors declare no conflicts.

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